PA194DHS: a new 194-state PA consistency machine

A complete corpus comparison, an exact construction, and a blank-start obstruction for the zero-spare transplant

Research prepared for Vladimir Reshetnikov

13 November 2026, Pacific time

Abstract

The requested sub-194 target was not attained. We supply the permitted alternative: an explicit, graph-distinct, binary one-tape machine with **194 working states**, together with an ordinary mathematical proof that PA proves its blank-tape nonhalting equivalent to Con(PA). PA194DHS combines the independently selected diagonal head $\Gamma(D)$ with visible-result, no-push modus ponens and a symmetric cached-head Shift implementation. Its command interface differs from both parent interfaces. Its allocation is $135 + 69 - 10 = 194$, and its literal 388-row table is included.

The comparison covers all 51 deliveries at the pinned repository revision: 47 filed research packets and four current incoming archives. A fresh literal-table census recognizes 34 distinct 194-state graphs, including auxiliary and rejected machines. PA194HS and PA194NS are exactly the same table; PA194N6 is also repeated in two current submissions. The new table matches none of the 34 graphs, even after control renaming, changing the graph-spanning start, reflection, or symbol interchange.

Two separately implemented physical checks verify the complete source interfaces. The independent literal-tape check covers all 4,752 primitive outcomes, arbitrary natural register values and an arbitrary nonempty spare suffix. The full blank tape reaches the exact clean source frame after 58,014 steps. A new 263,236-pair positivity invariant justifies the two specialized returns. As a supplementary negative result, the naive append-zero mutation of the primary PA194 cannot be rescued by choosing any of its 194 controls as start: four starts halt and all 190 others reach a provable blank ray within 77 transitions. None of these results is a state-count lower bound. A separate logical supplement shows that negative instances of diagonal vacuity suffice for the inherited PA refutation calibration; its direct implementations did not improve the state count. No new Lean/Rocq kernel theorem or completed iPRA consistency equivalence is claimed.

Contents

1 Result, conventions and evidence **3**
1.1 What is newly established . . . 3
1.2 Proof status . . . 4

2 Comparison of all supplied designs **4**
2.1 Pinned scope and reproducible inventory . . . 4
2.2 The substantive families . . . 5
2.3 Duplicates and invalid smaller-looking candidates . . . 6

3 Why PA194DHS is materially different **6**

4 Mathematical proof of the consistency equivalence **7**
4.1 The numerical calculus and exact command interface . . . 7
4.2 A uniform refutation compiler . . . 8
4.3 The two inserted zero tests and exact physical realization . . . 9
4.4 Consistency conclusion and proof boundary . . . 10

5 The inherited logical calibration, made explicit **10**
5.1 Formula coding and the PA-to-calculus direction . . . 10
5.2 Numerical refutations produce PA refutations . . . 11
5.3 Internal finite-trace conclusion . . . 12

6 Physical construction and independent verification **12**
6.1 Exact allocation and simultaneous folds . . . 12
6.2 Fresh positivity invariant . . . 13
6.3 Universal physical checks . . . 13
6.4 Complete blank initialization . . . 14

7 Search below 194 and a new all-start obstruction **14**
7.1 Physical contractions and decoder allocation . . . 14
7.2 The append-zero transplant cannot be fixed by changing its start . . . 15
7.3 Other source ideas and their limits . . . 16

8 A logical simplification: diagonal vacuity only for negations **16**
8.1 Positive vacuity at the same binder . . . 17
8.2 The two places in the inherited PA calibration . . . 18
8.3 Physical cost and the separate positive-offset experiment . . . 18

9 Reproduction, artifact map and attribution **19**

A Every supplied delivery **21**

B Every supplied 194-state graph class **23**

C Complete transition table of PA194DHS **25**

1 Result, conventions and evidence

The definitive machine is `machine/machine.tm`. Its working states are $0, \ldots, 193$, state 0 is initial, and 194 is a separate terminal halt label. The tape alphabet is $\{0, 1\}$, the tape is two-way infinite, and the initial tape is zero everywhere. The head begins at cell zero. Every defined transition writes one bit and moves exactly one cell left or right. The halt label has no transition rows and is excluded from the state count. All 194 working states are reachable in the directed control graph; this assertion does not mean that all are visited on the blank run.

Theorem 1.1 (Ordinary mathematical PA calibration)**.** *For the literal table supplied in this package,*

$$\boxed{\mathrm{PA} \vdash (\neg\, \mathrm{Halt}(\mathsf{PA194DHS}, \mathrm{blank}) \longleftrightarrow \mathrm{Con}(\mathrm{PA})).}$$

The proof combines a new exact command compiler and fresh physical certificates with the inherited ordinary separate-head PA calibration. The logical and implementation dependencies are stated below. This formulation does not assume Con(PA) in order to prove the equivalence. If PA is consistent, the machine does not halt; if a PA refutation exists, the compiled search halts. A bounded blank-tape run by itself establishes neither assertion.

The SHA-256 of the exact dense table, including its delivered newlines, is

`2c5385ddcb928af834c1b959058d5fdc9e01b1d2be3ee9cf7c50557e45bfb48e`.

The native presentation starts in 74, dispatches source frames in 84 and halts in 254. `machine/labels.json` gives the exact dense-to-native bijection. Dense and native presentations specify the same machine.

1.1 What is newly established

The positive contribution is a new combination with a different specified command interface, a new source specialization invariant, simultaneous physical folds, exact startup and complete independent physical replay. The individual diagonal-head, no-push and symmetric-Shift ideas are inherited. Their combination is not justified by assuming that the parents' physical invariants survive: they are recomputed for this table. In particular, the old Gamma-D positivity invariant has 263,244 pairs; the new one has 263,236.

The finite census and exact novelty check are fresh. The supplementary all-start obstruction strengthens the previously reported failure of the single native start 74 for the append-zero transplant. The searches below 194 are unsuccessful finite searches, described with their limits. The negative-only diagonal supplement is a further mathematical simplification of the searched calculus, with an independently reviewed proof and a separate, unsuccessful implementation-cost screen.

1.2 Proof status

Layer	Status and role
Literal table and state count	Complete, regenerated and independently parsed.
Source control flow	Two resolvers agree; an independent splice reconstructs the new source from PA194D.
Reachable specializations	A finite inductive positivity relation is computed afresh and checked for closure and step agreement.
Physical implementation	Complete symbolic contracts over arbitrary registers; four list lemmas are derived from the actual table. Exact startup is checked separately.
Command refutation compiler	Explicit all-store word identities and a structural proof; free-term symbolic checks corroborate them.
Logical calibration to PA	The inherited ordinary syntactic proof transformations are retained, explained and independently reviewed.
Proof-assistant endpoint	No new end-to-end Lean/Rocq theorem; the repository's stronger iPRA goal remains open.

Executable certificates are mathematical evidence with a stated checker trust boundary; they are not silently promoted to a proof-assistant kernel.

2 Comparison of all supplied designs

2.1 Pinned scope and reproducible inventory

The comparison is pinned to ProveIt commit `3cc7fbf75ba111a0da696ef2f6001dd528891bd5`. The two requested input locations are `Computability/BusyBeaver/Research/PA` and `docs/incoming`. The archived packet ledger contains 47 filed research deliveries. Four new ZIPs remain in `docs/incoming`: the HS, N5/N6, N6-research and NS packets. The report compares their machine families, source changes, proof interfaces and rejected attempts. Appendix A lists every delivery.

The literal-table scan examines numeric five-column binary tables, including nested archives. It does not claim to recognize arbitrary machines hidden only in executable source or other encodings. The manuscript and source review complements this declared syntactic scope.

Fresh census quantity	Count
Filed packets plus current incoming ZIPs	51
Recognized literal-table occurrences	1,811
Distinct exact byte forms	683
Graph classes admitting a spanning start	487
Distinct 194-state graph classes	34
Disconnected synthetic byte forms excluded from that graph count	2

The older repository ledger reports 3,803 table occurrences before duplicate table files were removed during filing. That historical number and the fresh 1,811-occurrence count describe different file populations. Neither counts validated consistency machines. Among the 34 frontier classes, the audit classifies 21 as reported constructions, nine as auxiliary research tables and four as explicitly rejected designs. These labels describe the supplied evidence; this report does not freshly prove all historical constructions.

2.2 The substantive families

Family	States	Main distinction and comparison result
Unary/U209/U208	209–208	Unary-power/pointed-separation search; entry-context and history folds. Several separately titled U208 packets repeat the same graphs. Useful engine and proof-compiler ancestors, not competitive with the current frontier.
PA209/PA196	209, 196	James Chang's upstream source, numerical calculus and register engine. PA196 supplies the common physical and logical framework. Its documented formal PA endpoint includes nonhalting $\Rightarrow$ Con(PA), not both machine-specific iPRA reductions.
PA195 and variants	195	Original, separate-head, cached, scanner and combined-gate allocations. They expose the source-versus-physical-cost distinction: shorter source does not automatically mean fewer controls.
Primary PA194	194	Same 3,673-site separate-head source as its parent, compressed to 67 raw AX readers and seven compatible history folds, with 134 fixed controls. An A/D allocation companion changes register placement.
Cached C/CS/CZ	194	Cache-centered command execution; symmetric equality and known-zero returns. Five principal C submissions are one graph. PA194X and PA194CSZ also coincide.
Radix 4R/5/6	194	Different arithmetic formula-code construction or association. The unused residues in larger radices require an explicit calibration; arithmetic cost and reader sharing must be assessed jointly.
J and ZR	194	Changed return conventions justified by known-zero source invariants. Some alternative return registers instead cause startup halting, so runtime-only checks are insufficient.
D	194	Emits $\Gamma(D)$ independently of the separation parameter, with a new diagonal selector and two reachable zero-test specializations. Its allocation is $134 + 69 - 9$.
Q	194	Exchanges the curried quantifier-premise order; uses explicit proof translations and a changed emitter.
R and associated layouts	194	Different decoder associations, entry/close conventions and auxiliary layouts. The comparison keeps exact graph identities separate from their suggested semantics.
N, N5, N6	194	Successful MP returns the visible conclusion and omits the dummy stack push. N5/N6 combine this protocol with radix recodings; the two current N6 copies are identical.
H and HS/NS	194	No-push visible-result MP combined with a cached-head-only Shift test. HS/NS use a symmetric comparison. HS and NS are byte-identical despite separate reports.

Family	States	Main distinction and comparison result
PA194DHS (new)	194	$\Gamma(D)$ emitter combined with the HS command interface. It requires a changed diagonal selector, fresh zero invariant and ten simultaneous folds. It is absent from every supplied frontier graph.

The complete 34-class table with file identities and evidence categories is Appendix B; machine-readable records and representative tables are in `comparison/`. The nine auxiliary classes are retained in the novelty screen, so a result is not called new merely because an older table was not a packet's headline machine.

2.3 Duplicates and invalid smaller-looking candidates

PA194HS and PA194NS have exact table digest

`19935635f2c5722f5a72977d20a7019764a28573e12d1d0bce398245576c33b1`.

The N6 table repeated in the two-machine and N6 research packets has digest

`36da42a81d605ccb8b5edae197c7cde311447bd95b9bf1ab771a3e726a41c6f4`.

Thus the four current ZIP deliveries add four distinct frontier graphs, not one new graph per machine name or report. Five main C manuscripts likewise contribute one principal graph.

Three rejected comparator-return tables were independently replayed and each halted on the blank tape after exactly 1,361 transitions. These are not candidates calibrated to PA by the cited source-start argument. A fourth rejected frontier class uses source addresses that violate its seven-zero scanner. The input census intentionally retains all four as negative controls and distinguishes them from accepted constructions.

3 Why PA194DHS is materially different

Let $p(a,b) = (a+b)(a+b+1)/2+b$, $I(a,b) = 4p(a,b)$, $U(i,B) = 4p(B,i)+2$, and $\Gamma(B) = I(B,U(B,B))$. The store is (t,w,d,h): visible theorem, Cantor-coded stack, numerical offset and cached theorem. The full semantics are proved in the next section. Two contrasts locate the new design precisely.

First, the new successful MP leaves the conclusion in both the visible theorem and offset, while clearing the cache. The cached Gamma-D parent leaves that conclusion in the cache and clears the offset. For example, $I(0,2) = 20$, so

$$\mathsf{MP}_2(0,w,0,20) = (2,w,2,0)$$

in PA194DHS; the Gamma-D cached parent returns $(0,w,0,2)$. The new compiler therefore uses a final Pop, with a reserved zero cell, where the parent's compiler could use a Shift.

Second, the new AX diagonal follows $D = d + x$, while HS/NS follows the independent input $A = c$. On the zero store, $\mathsf{AX}_{0,1,0}$ therefore pushes $\Gamma(1) = 832$ first in PA194DHS, versus $\Gamma(0) = 20$ in HS/NS. The other heads keep their former parameter assignments. This difference cannot be explained by globally swapping parameter or register names. After the successful MP_2 example and an AX call with all three explicit parameters zero, the new diagonal is $\Gamma(2) = 5712$; both parent protocols emit $\Gamma(0) = 20$ in that same two-command experiment.

The new diagonal selector is correspondingly

$$\mathsf{MP}_0; \mathsf{AX}^D_{0,B,0}; S^2; \mathsf{Pop}^8.$$

It selects $\Gamma(B)$ independently of the guarded separation head. The table also has a different complete control graph. The novelty checker compares full canonical transition tuples, not only digests. It finds no match against any of the 34 supplied graphs after arbitrary state renaming, every graph-spanning start, global left/right reflection and global symbol interchange. The last comparison is stronger than the blank-preserving convention requires, because the designated blank symbol is fixed at zero. This is novelty relative to the pinned supplied corpus, not a claim about all unpublished machines or every future repository revision.

4 Mathematical proof of the consistency equivalence

4.1 The numerical calculus and exact command interface

Let $p(a,b) = (a+b)(a+b+1)/2 + b$, write $[a;w] = p(a,w)$, and put

$$I(a,b) = 4p(a,b), \quad M(a,b) = I(a,b)+1, \quad U(a,B) = I(B,a)+2, \quad E(a,b) = I(a,b)+3.$$

Code zero is truth and code one is contradiction. Negation is $N(B) = I(B,1)$; conjunction and existential quantification are the standard classical abbreviations. Cantor stacks have total projections: $\mathrm{hd}(0) = \mathrm{tl}(0) = 0$ and $[0;0] = 0$.

The unchanged numerical calculus has modus ponens, generalization of theorems $B \mapsto U(i,B)$, every $\Gamma(B) = I(B, U(B,B))$, and the heads

$$\begin{aligned}
h_1(D,H) &= (\exists x_D U(D,H)) \to H,\\
h_2(D,A,H) &= U(D,U(A,H)) \to U(A,U(D,H)),\\
h_3(D,A,H) &= M(H,D) \to (E(H,A) \to M(A,D)),\\
h_4(D,A,H) &= E(H,D) \to (M(A,H) \to M(A,D)),\\
h_5(D,H) &= \exists x_D E(H,D),\\
h_6(D,A,H) &= U(D,A) \to (U(D,A\to H) \to U(D,H)),\\
\rho(D,A,H,d) &= ((H\to d)\to D)\to((D\to H)\to(A\to H)) \quad (d \le D).
\end{aligned}$$

Its guarded pointed-separation formula is

$$\begin{aligned}
V(a,b,c) &= M(a,b) \to U(c, M(a,b)), & P(a,b,c) &= c \to M(c,b),\\
S_*(a,b,c) &= U\big(b, (M(a,b)\to P(a,b,c)) \wedge (M(c,b)\to c)\big),\\
Q_*(a,b,c) &= U(c, S_*(a,b,c) \to N(P(a,b,c))),\\
G(a,b,c) &= N(V(a,b,c) \to Q_*(a,b,c)).
\end{aligned}$$

Here G is an axiom when $c \neq a,b$; put $g = G$ under that guard and $g = 0$ otherwise. The code-freshness fact that all labels occurring in a formula code B are smaller than B justifies $\Gamma(B)$.

A command store is (t,w,d,h), comprising top, stack, offset and cache. The equations below hold at the inherited command boundaries, where the scratch register X is zero. They allow arbitrary natural values of the four displayed store components, not arbitrary unclean physical scratch. For $b = d + x$, the new source has

$$\mathsf{MP}_x(t,w,d,h) = \begin{cases} (b,w,b,0), & I(t,b) = h,\\ (0,w,b,0), & I(t,b) \neq h, \end{cases}$$

$$S(t,w,d,h)=\begin{cases}\mathsf{halt}, & h=1,\\ (h,\mathrm{tl}(w),d,\mathrm{hd}(w)), & h\neq 1.\end{cases}$$

For $D=d+x$, $H=h+y$, axiom generation is

$$\mathsf{AX}^{D}_{A,x,y}(t,w,d,h)=\big(g(H,D,A),[q_1,\ldots,q_9,t;w],0,0\big),$$

where

$$(q_1,\ldots,q_9)=(\Gamma(D),h_6,h_5,h_4,h_3,h_2,h_1,\rho(D,A,H,d),U(d,h)).$$

The free-constructor audit checks every displayed formula tree and stack position. All $q_1,\ldots,q_8,g$ have residue zero modulo four; $q_9\equiv 2 \pmod 4$. None equals one, without any consistency assumption.

For an explicit distinction, $I(0,2)=20$, and $\mathsf{MP}_2(0,0,0,20)=(2,0,2,0)$. A subsequent $\mathsf{AX}^{D}_{0,0,0}$ emits $\Gamma(2)=5712$. The hybrid Gamma-A parent emits $\Gamma(0)=20$ in this situation, while the cached Gamma-D parent's MP puts the consequent in h and clears d. Thus the new correlation is not a renaming of the command parameters.

4.2 A uniform refutation compiler

Write $A_0=\mathsf{AX}^{D}_{0,0,0}$, and execute words left to right. Define

$$\mathsf{Push}=A_0;S^9;\mathsf{MP}_0,\quad \mathsf{Pop}=A_0;S^{11},\quad \mathsf{Discard}=\mathsf{MP}_0.$$

Direct use of the stack equations gives

$$\begin{aligned}\mathsf{AX}^{D};S^9(t,w,d,h)&=(q_8,[t;w],0,U(d,h)),\\ \mathsf{AX}^{D};S^{10}(t,w,d,h)&=(U(d,h),w,0,t),\\ \mathsf{Push}(t,w,d,h)&=(0,[t;w],0,0),\\ \mathsf{Discard}(t,w,0,h)&=(0,w,0,0),\\ \mathsf{Pop}(t,[a;w],d,h)&=\begin{cases}\mathsf{halt}, & t=1,\\ (t,w,0,a), & t\neq 1.\end{cases}\end{aligned}$$

The first two identities are total: their tested caches are zero or non-one generated heads. Push's final MP fails by the residue difference between implication and universal codes. Discard follows because both MP branches agree when the prospective conclusion is zero. These arguments include the aliases $[0;0]=0$ and $I(0,0)=0$.

For $1\le j\le 7$, from $(t,w,0,0)$ the selector

$$\mathsf{Discard};\mathsf{AX}^{D}_{A,D,H};S^{j+1};\mathsf{Pop}^{9-j}$$

returns $(q_j,w,0,0)$. The Gamma-D case is specifically

$$\mathsf{Discard};\mathsf{AX}^{D}_{0,B,0};S^2;\mathsf{Pop}^8,$$

which selects $\Gamma(B)$ for every B. The dominated head, for $u\le a$, is selected by

$$\mathsf{Discard};\mathsf{MP}_u;\mathsf{AX}^{D}_{b,a-u,c};S^9;\mathsf{Pop};$$

the gate is selected by

$$\mathsf{Discard};\mathsf{AX}^{D}_{A,D,H};\mathsf{Pop}^{10}.$$

Generalization is

$$\mathsf{Gen}_i = \mathsf{Push}; \mathsf{MP}_i; S; \mathsf{AX}^D_{i,0,0}; S^{10}.$$

It sends $(a, w, 0, 0)$ to $(U(i, a), w, 0, 0)$ even when $a = 1$.

For each numerical derivation p, construct a word $C(p)$ satisfying: from every $(t, w, 0, 0)$ it either halts or returns $(\mathrm{concl}(p), w, 0, 0)$. Use the displayed selectors for axioms, and append Gen_i at generalization. For MP with premise proofs $p_0 : a$ and $p_1 : I(a, b)$ use

$$\mathsf{Discard}; \mathsf{Push}; C(p_1); \mathsf{Push}; C(p_0); \mathsf{Push}; S; S; \mathsf{MP}_b; \mathsf{Pop}.$$

The reserved zero gives $(0, [a, I(a, b), 0; w], 0, 0)$ before the two Shifts. The second Shift halts if $a = 1$; otherwise MP returns $(b, [0; w], b, 0)$. The final Pop halts if $b = 1$; otherwise it consumes the reserved zero and returns $(b, w, 0, 0)$. This proves the invariant by induction on the finite proof tree. In particular a proof of one yields $A_0; C(p); \mathsf{Pop}$, which halts from every incoming store. The all-store qualification is essential because the scheduler retains its store between attempted words.

The unchanged scheduler enumerates all finite command words with positive continuation codes. Specifically, for suffix code z,

$$e(S, z) = p(0, z), \quad e(\mathsf{MP}_x, z) = p(1, p(x, z)), \quad e(\mathsf{AX}_{A,x,y}, z) = p(A + 2, p(y, p(x, z))).$$

The list encoder uses $E(\epsilon) = 0$, $E(c :: w) = 1 + e(c, E(w))$. Positive continuations decrease, so each nonhalting candidate finishes; fresh candidates run through consecutive integers from two. Prefixing the universally halting word by arbitrary AX commands gives unbounded fresh candidate codes and avoids the omitted initial values. All proof-word constructors, bounded candidate traversals and finite trace concatenations are primitive recursive, and PA verifies their correctness by ordinary induction.

Conversely, annotate t, h and every canonical stack entry with a numerical-calculus proof. Zero carries its fixed truth proof; numerical zero-stack padding is treated explicitly. AX emits axioms, a generalization of the old cache, and the old top. A successful MP produces its conclusion; a failed MP produces truth. Shift transports proofs and halts only on a cached proof of one. The offset needs no proof annotation. This is a syntactic invariant, not an appeal to arithmetic reflection.

4.3 The two inserted zero tests and exact physical realization

The command argument concerns the ordinary source with unconditional returns. The literal implementation replaces its site-26 jump by a zero test on X and its site-3553 final AX jump by a zero test on D. The positivity abstraction records for each of the nine registers whether it is zero or positive. Its recomputed least closure has 263,236 location/mask pairs. Both ordinary and specialized programs have this same closed set and identical successor sets there; the tested register is zero in every mask at each changed site. The finite checker verifies these facts. Induction on executions therefore proves exact equality of the two seeded source runs. Reusing the older Gamma-D closure would have been unjustified: its 263,244-pair set is different.

The new source has 3,650 sites and 4,752 decrement-branch-inclusive outcomes. The independent literal checker reconstructs its sites from the pinned Gamma-D source by explicit address splices, compares two separate control-flow resolvers, rederives four tape-list lemmas from the literal table, and proves every source outcome for arbitrary natural register values, arbitrary natural header gap and every nonempty finite spare list. It uses no arithmetic-box oracle. The ten simultaneous history folds pass forty local two-cell contracts with arbitrary exterior tape. Every source simulation lasts strictly positive time and preserves the complete frame, including spare-list effects.

Exact blank execution reaches native state 84, head -2, source site 25, logical registers $(2, 0, 0, 0, 0, 0, 0, 0, 0)$ and 26 spare registers of value one after 58,014 transitions, with no earlier halt. Independent set and bytearray simulations compare the entire tape and zero exterior. Concatenating positive simulations transports source halts to tape halts. If the tape first halted after T transitions while the first $T + 1$ source steps were nonhalting, their simulations would produce the same deterministic physical run for more than T nonhalting transitions, a contradiction. This supplies both finite-trace implications in PA.

4.4 Consistency conclusion and proof boundary

The supplied separate-head calibration provides primitive-recursive translations between PA refutations and refutations of the numerical calculus, with correctness provable in PA. It uses diagonal-code freshness, reverse membership rewriting, neighborhood equality, and arithmetic finite-comprehension proof templates. The present construction retains that calculus exactly; Gamma-D changes only the correlated command parameters used to select its already available diagonal instances. Composing that calibration, the new command compiler and annotation invariant, the fresh zero-test proof and the literal simulations gives

$$\mathrm{PA} \vdash \big(\neg\,\mathrm{Halt}(\mathrm{PA194DHS}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})\big).$$

This is an ordinary mathematical PA-provability argument with executable finite and universal symbolic certificates. The inherited logical calibration is an explicit dependency; these certificates are not a new end-to-end Lean or Rocq theorem. Standard-model nonhalting remains conditional on the consistency of PA.

5 The inherited logical calibration, made explicit

The construction above changes the command protocol, not the numerical calculus. The substantive inherited theorem is the existence of primitive-recursive transformations in both directions between PA refutations and numerical proofs of code one, whose correctness PA proves. The supplied calibration texts are retained byte-for-byte under `references/logical/`; this section explains the essential arguments and the interfaces on which the transfer depends [6, 2, 3]. It is not an assertion that the upstream formal project already exports both directions for this literal machine.

5.1 Formula coding and the PA-to-calculus direction

For each nonzero formula constructor, its relevant subcodes and variable labels are strictly smaller than the constructed code. Strong induction on codes proves that variable x_B does not occur in the formula coded by B. The exceptional numerical identity $I(0, 0) = 0$ is treated as truth, not as a well-founded recursive decomposition of zero. This yields the diagonal schema $B \to U(B, B)$, including its separate truth-code case.

The previous bundled diagonal gate is

$$F_\Gamma(a, b, c) = N\big(\Gamma(c) \to (V(a, b, c) \to Q_*(a, b, c))\big).$$

Classical implicational templates convert a bundled instance into $\Gamma(c)$ and $G(a, b, c)$, and convert those two heads back into the bundle. Each $\Gamma(B)$ is obtainable using a bundled instance with $c = B$ and $a = b = B + 1$. Thus separating the diagonal changes the number and order of pushed heads but

preserves a primitive-recursive proof translation. The ten-item command selectors in the present proof account for that changed stack protocol explicitly.

The inherited PA-refutation compiler needs two-direction membership rewriting, a neighborhood-equality implication and an equality witness. The point of the diagonal replacement is that it recovers these interfaces without reinstating the discarded equality-vacuity axiom. Here is the critical reverse-rewriting construction.

For one of the required membership substitutions, put

$$C = E(a,b) \to (\chi(b) \to \chi(a)), \qquad \varphi = N(C), \qquad c = \ulcorner \varphi \urcorner.$$

The predicate χ is a membership atom with its argument in either slot, including the diagonal case $M(t,t)$. For distinct original labels, the strict code bounds make c fresh. Pointed separation at (a,a,c) supplies $N(Q_*)$. The second part of its matrix, together with the propositional consequence $\varphi \to N(\chi(a))$, yields

$$S_* \to U(a, M(c,a) \to N(\chi(a))).$$

The retained equality witness, direct membership rewriting and membership-only vacuity transport this to

$$S_* \to (M(c,b) \to N(\chi(b))).$$

Under φ, the point condition supplies $M(c,a)$, direct rewriting supplies $M(c,b)$, and φ also supplies $\chi(b)$. Discharging the assumptions gives $\varphi \to (S_* \to N(P))$. Generalization at c and the available diagonal instance $\Gamma(\varphi)$ give $\varphi \to Q_*$, contradicting $N(Q_*)$. Double-negation elimination therefore gives C. Coincident original labels are handled by propositional reasoning and the diagonal-membership case, rather than by assuming the fresh-label argument still applies unchanged.

For neighborhood equality, begin with the fresh binder $z_0 = \ulcorner E(a,b) \urcorner$. The instance $\Gamma(E(a,b))$, universal distribution and the two membership rewrites give the neighborhood implication at z_0. The equality witness and membership vacuity transport the binder to each required $z \neq a, b$, also supplying the neighborhood witness. These replace the equality-vacuity uses in the original refutation compiler. The audit checked these interfaces against the upstream `Logic/Reverse.lean`, `Neighborhood.lean` and `Vacuity.lean` dependencies, rather than assuming that a standard-model interpretation alone gives an internal PA theorem.

All substitutions, fresh-code construction, fixed propositional templates and finite derivation concatenations are operations on syntax. They have primitive-recursive implementations, and the inherited calibration proves their correctness by arithmetic induction. This direction supplies the numerical proof of one required by the new all-store command compiler.

5.2 Numerical refutations produce PA refutations

Interpret membership by finite binary membership,

$$\mathrm{Bit}(s,z) \quad \Longleftrightarrow \quad \exists u, v\, [s = (2u+1)2^z + v \ \wedge \ v < 2^z],$$

equality by ordinary equality, and $U(a,B)$ by universal quantification over the translation of B. PA proves $\mathrm{Bit}(s,z) \to z < s$, so it proves the negation of the translation of numerical code one, $M(0,0)$: the two zeros here are the same variable label, giving $\mathrm{Bit}(x_0,x_0)$. Code zero receives a fixed truth translation and a fixed proof relating it to the alias $I(0,0)$.

For every arithmetic formula $\theta(z,\vec{u})$, PA proves finite binary comprehension:

$$\forall n\, \exists s < 2^n\, \forall z < n\, (\mathrm{Bit}(s,z) \leftrightarrow \theta(z,\vec{u})).$$

The induction step keeps the old code when $\neg\theta(n,\vec{u})$ and adds 2^n when $\theta(n,\vec{u})$. For the internal syntactic statement, one substitutes the code of θ into this fixed induction-proof template. PA verifies that the resulting finite string is a PA proof. This does not assert a uniform arithmetical truth predicate for all formulas and uses no reflection principle for PA.

In the pointed-separation translation with $a \neq b$, apply this template to

$$(\mathrm{Bit}(x_a, z) \vee z = x_b) \wedge T(c)[z/x_b]$$

with a bound such as $x_a + x_b + 1$. This finite set includes the required point if its predicate holds, includes all relevant members satisfying the predicate, and contains only elements satisfying the predicate. For $a = b$, the inner binder changes both occurrences of that variable label. The correct matrix is instead

$$(z = x_b) \wedge T(c)[z/x_b].$$

Inside the binder the other antecedent is $\mathrm{Bit}(z, z)$, which PA refutes. This case distinction is essential: treating the first variable as free in the collision case would be an incorrect soundness proof. The separator label is fresh by the code bound, and the guard $c \neq a, b$ supplies the remaining freshness.

The diagonal and other logical heads have uniformly generated first-order or propositional PA proof templates. The dominated implicational head is arithmetically interpreted by its propositional template; completeness of the restricted supply is part of the inherited numerical proof calculus. Each numerical proof inference is translated by a syntactic proof operation. A numerical proof of code one thereby produces a PA proof of its translation and, using the fixed negation proof above, a PA refutation.

5.3 Internal finite-trace conclusion

Writing H_R for halting of the new seeded register source, the two syntactic translations, the all-store compiler, cofinal scheduler and proof-annotation invariant give

$$\mathrm{PA} \vdash (H_R \longleftrightarrow \exists p\ \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner))\,.$$

The fresh ordinary/specialized source equivalence and the literal positive-duration simulation then replace H_R by halting of PA194DHS. Negating the existential refutation statement proves the displayed nonhalting/consistency equivalence. The missing upstream end-to-end iPRA endpoint is neither used as a theorem nor inferred from the Python certificates.

6 Physical construction and independent verification

The complete source tree has 3,650 primitive occurrences. A decrement has two possible outcomes, producing 4,752 source contracts in total. The source-to-tape representation carries the binary source address, an affine zero gap, nine unary primary registers and a nonempty finite list of spare registers. Logical register order is $(N, T, W, A, D, H, X, Y, L)$; physical positions are $(6, 2, 0, 4, 3, 7, 8, 5, 1)$, numbered from zero.

6.1 Exact allocation and simultaneous folds

The raw machine has 204 working controls: 135 fixed controls and 69 AX reader controls. Ten disjoint pairs are identified, giving

$$135 + 69 - 10 = 194.$$

For each pair, the incoming transition to one member establishes previous bit zero and to the other establishes previous bit one. The replacement uses that retained tape bit to distinguish the two former controls. For both histories and both current read bits, the checker runs the final simultaneously patched table to the exact old target, head position and two-cell tape. Forty local cases certify the ten folds; no exterior cell is visited during the local detours. Full physical source contracts then check the folded table itself. The fold list and all finite local traces are in `machine/folds.json` and `evidence/local_cases.json`.

6.2 Fresh positivity invariant

The ordinary source uses two unconditional returns. The physical source replaces them with decrements whose zero branch has the same successor: site 26 tests scratch register X (logical 6), and site 3553 tests D (logical 4). The source's MP change retains a different offset, so an old Gamma-D invariant is not an admissible substitute for a fresh check.

A mask $m < 512$ records exactly which of nine registers are positive. Increment sets its bit. A decrement on a zero bit follows its zero target; a decrement on a positive bit follows its positive target and nondeterministically keeps or clears that bit. This is a sound finite abstraction of all natural register values. Starting at $(25, 1)$, both the ordinary and specialized programs have the same closed relation of **263,236 pairs** (pc, m). Every abstract successor agrees between the two programs throughout this relation. At the two changed sites, the tested bit is always zero.

Induction on the source run therefore proves equality of the concrete ordinary and specialized executions from the clean seed. This is a proof over arbitrary run lengths, not a bounded simulation. The finite relation, its hash, masks at changed sites and closure certificate are regenerated by `build.py`. The independent source splice and two control-flow resolvers additionally check that the delivered instruction file really is the intended modified source.

6.3 Universal physical checks

`build.py` checks the contracts using the inherited register-box interface. `verification/check.py` uses a separate literal-tape engine: it imports neither the construction compiler nor that arithmetic oracle. Its four generic finite-list sweep lemmas are rederived from the actual transition rows by an arbitrary one-block calculation and list induction. Each primitive outcome is then verified on the entire symbolic tape with nine independent natural register values, both decrement branches, an arbitrary natural extra gap and an arbitrary nonempty finite spare list of arbitrary natural values. The expected appended spare is one-valued; that resource policy is part of the contract.

Verification quantity	Result
Working states	194
Defined binary transitions	388
Primitive source sites	3,650
All-natural source outcomes	4,752
Literal list lemmas rederived	4
Independent symbolic operations	1,096,025
Largest symbolic operation count for one outcome	385
Smallest constant in a physical duration	3
Positivity relation pairs	263,236
Exact blank-start transitions	58,014

Every accepted nonhalting source segment has a positive physical duration, expressed by an affine formula with nonnegative coefficients. List-sweep and unary-sweep rules carry their induction proofs; a long physical loop is never replaced merely by a finite sample. This positivity and complete termination of each segment support both directions of the finite-trace equivalence described in the mathematical proof.

6.4 Complete blank initialization

The native start 74, head zero and empty set of one-cells reach dispatcher 84 at head -2 after **58,014 transitions**, without an earlier halt. The source counter is 25, its path is `100000`, and the logical register vector is $(2, 0, 0, 0, 0, 0, 0, 0, 0)$. The header distance is 48 and the complete suffix contains 26 one-valued spares. Both a set-based simulator and an independent bytearray simulator compare the whole finite-support tape with this expected frame. Equality on a selected window would not establish this initialization claim.

7 Search below 194 and a new all-start obstruction

No smaller machine is certified by this work. The finite experiments below are recorded to make clear what was attempted and what remains open. An optimization score, a source contraction, or a table filename containing `193` is not counted as a successful consistency machine.

7.1 Physical contractions and decoder allocation

The physical screens operate on explicit literal tables. They try common replacement actions for two controls, or separately redirected incoming edges of an eliminated control. The replacement must reproduce the specified tape, target control and head position inside the tested window, and must avoid circular reliance on rows being rewritten. Passing one representative window would not establish a global proof; all required abstract contexts must pass, followed by the complete source and startup checks. No proposed contraction passed the screens below.

Input / rule	Radius	Step bound	Context scope
Primary PA194, pair merger	4	80	All-tape abstract contexts.
Primary PA194, pair merger	6	120	Blank-start abstract contexts.
Top-only Shift, pair merger	6	300	Blank-start abstract contexts; this input has 195 controls.
PA194DHS, pair merger	7	300	Blank-start abstract contexts; all working endpoints considered.
PA194DHS, incoming-edge bypass	7	300	All-tape abstract contexts; exact old two-step endpoint.
PA194DHS, joining-prefix pair	4	40 / 120	Blank-start abstract contexts; a later old configuration may be the common endpoint.
PA194DHS, joining-prefix pair	6	80 / 240	274,589 abstract state/window pairs.
Top-only Shift, joining-prefix pair	6	80 / 240	268,907 abstract state/window pairs.

Here a radius-r window has $2r + 1$ cells. These abstractions and bounded continuations give sufficient local rewrite rules, not complete searches through halting-equivalent machines. A failed overapproximation can hide a reduction that requires a stronger global invariant. For a joining-prefix screen, the two bounds concern the old prefix and replacement continuation, respectively. Both paths must avoid rewritten controls before their common endpoint. These three additional screens also returned zero candidates.

A separate 15,120-case finite screen permutes all seven generated head blocks in the DHS association and independently chooses the diagonal argument from A, D, H. Its best allocation score is 194. Source semantics, structured returns and simultaneous local folds still require their own checks whenever an allocation score looks promising.

Three seeded grammar searches, including folds with eligible fixed passive readers, likewise supplied no certified sub-194 table:

Search	Seconds	Proposal slots	Best score
DHS shared-subtree reassociation	600	11,362,240	194
DHS head order, parameters and reassociation	600	8,606,089	194
Normalized top-only Shift reassociation	120	2,243,386	195

The seeds are 202611141931, 202611141932 and 202611141933, respectively. The total of 22,211,715 counts proposal slots, including skipped mutations, repeated trees and rejected proposals; it is not a count of distinct machines or proofs. The corresponding accepted-mutation counts are 547,685, 3,012,683 and 130,995. The optional bundle at `search/local_and_grammar/` records exact inputs, seeds, stopping budgets, observations and limitations. Discovery is not required by `verify_all.py`.

An early top-only mutation experiment had inconsistent indexing between opaque and expanded increment macros. That run is excluded from trusted search accounting. A corrected normalized reassociation experiment is recorded separately. The accepted PA194DHS construction and its complete verification do not depend on the rejected search run.

7.2 The append-zero transplant cannot be fixed by changing its start

The PA engine appends one-valued spares. The ZFC work motivates trying zero-valued spares; this changes a resource convention and requires a new complete startup proof. Define Z to be the primary PA194 table in native labels, with exactly one transition changed:

$$(198, 0): \quad (1, L, 40) \longmapsto (0, L, 40).$$

All other transitions remain exactly those of the primary PA194 table, including every control retained in that table. There are still 194 working controls. Earlier work already rejected its native start 74. We check all possible working-state starts, with the blank tape and head zero held fixed.

Proposition 7.1 (All-start obstruction for this exact transplant)**.** *The blank run of Z started at native controls $0, 66, 94, 97$ halts after 37, 33, 34, 31 transitions respectively. At each of the other 190 possible starting controls, the run enters a certified nonhalting blank ray within 77 transitions. Thus choosing another existing initial control cannot turn this exact mutation into the intended PA proof search.*

Proof. The supplied verifier calculates the exact finite prefix for each start. At a reported ray endpoint, the head reads zero, the transition has the form $(q, 0) \mapsto (w, d, q)$, and the entire half-line ahead of

the head in direction d is zero. Induction gives the same transition forever: each step enters a fresh zero cell, leaves any written bit behind, and never changes state. The finite set of one-cells certifies the half-line premise exactly. In the four halting cases, the finite prefix reaches the unique terminal control 254. The complete case split contains every working control and no other control. □

For the original start 74, the exact endpoint after 76 steps is state 35, head 4 and one-cells $\{-3, 1\}$. Its transition $35/0 = 1R35$ gives the rightward ray. The maximum 77-step bound concerns entry into a certificate, not the execution time of a nonhalting machine.

This proposition is stronger than failure to find a good starting state within a long numerical timeout. It classifies every start by a finite proof of its subsequent behavior. It is specific to Z: it does not exclude zero-spare designs with further table or source changes, and it does not establish any lower bound for the requested problem. For the 190 ray starts, PA itself proves nonhalting. Assuming PA is consistent, a PA proof of the requested equivalence for one of these starts would therefore give a PA proof of Con(PA), contrary to the second incompleteness theorem. The four other starts actually halt, so under Con(PA) they fail the intended standard-model equivalence. These are distinct conclusions: mere consistency is not used as a blanket soundness assumption for every sentence PA might prove.

7.3 Other source ideas and their limits

A 105-way exploratory cross screen combines three source-emitter grammars, seven command-controller choices and five arithmetic coefficient layouts. No screened count falls below 194. In particular, combining radix 5 or 6 with the symmetric cached controller costs an extra state in the tested layouts because two prospective folds compete for the same endpoint. This is why reductions from different parents cannot simply be added.

We also examined restricting pointed separation to a stronger guard $c > \max(a, b)$, hoping to replace two equality checks by a one-sided test. The existing gate also supplies membership vacuity. The reverse-rewrite proof needs vacuity at an original small label for a membership atom containing the much larger fresh separator label. The stronger guard removes those required instances. Consequently the current PA calibration does not transfer to that proposed shortcut. A new logical proof or a separate vacuity interface would be required; no smaller calibrated machine is claimed on this basis.

ConciseIPRA supports the methodological use of explicit proof translations and compact derived interfaces, but replacing the searched calculus by iPRA would change the consistency target unless a PA calibration is proved. The ZFC work supplies useful command, spare and startup ideas; its calibration cannot be substituted for a PA calibration. The successful construction preserves the required PA target throughout.

8 A logical simplification: diagonal vacuity only for negations

This supplement concerns the numerical calculus; it does not alter the accepted PA194DHS transition table. Replace the full diagonal schema

$$\Gamma(B) = B \to U(B, B)$$

by its negative instances alone,

$$\Gamma^{-}(B) = \Gamma(N(B)) = N(B) \to U(N(B), N(B)), \qquad N(B) = B \to 1.$$

Keep all six ordinary heads, the dominated implicational head, the membership-vacuity/pointed-separation gate and its original distinctness guard, modus ponens, and theorem generalization. Call this

calculus $\mathcal{C}_-$. All its displayed diagonal instances remain sound under the inherited PA interpretation, by the same code-freshness lemma.

8.1 Positive vacuity at the same binder

For a fixed numerical label z, write $\Box A = U(z, A)$ and $\Diamond A = N(\Box N(A))$. These are abbreviations for the original numerical constructors. We use the retained heads

$$\Diamond\Box A \to A, \qquad \Box A \to (\Box(A \to B) \to \Box B),$$

closed generalization, and the derived classical implicational proof operations. The latter include contraposition, composition, and both double-negation implications; their derivations use the dominated basis and the first ordinary head, and use no diagonal or equality-vacuity axiom.

Vacuity-negation lemma. From a closed derivation of $A \to \Box A$ one can construct, uniformly, a closed derivation of $N(A) \to \Box N(A)$. Indeed contraposition gives $N(\Box A) \to N(A)$. Generalize this closed theorem and apply the sixth head to obtain

$$\Box N(\Box A) \to \Box N(A).$$

The first head is $N(\Box N(\Box A)) \to A$. Contraposition followed by double-negation elimination gives

$$N(A) \to \Box N(\Box A).$$

Composition proves the assertion. Generalization is applied only to closed theorems; there is no assumption-dependent generalization step. This is the proof operation named `negationVacuity` in the supplied `PA196/Logic/Vacuity.lean`.

Positive-from-negative lemma. For arbitrary B, z, a closed derivation

$$N(B) \to U(z, N(B))$$

uniformly yields a closed derivation $B \to U(z, B)$. Apply the preceding lemma with $A = N(B)$, and compose the three implications

$$B \;\longrightarrow\; N(N(B)) \;\longrightarrow\; U(z, N(N(B))) \;\longrightarrow\; U(z, B).$$

The first arrow is double-negation introduction. The middle arrow is vacuity-negation applied to the input proof. The last arrow is obtained by generalizing the closed double-negation-elimination theorem $N(N(B)) \to B$ and applying universal distribution. Thus replacement under U is justified by an explicit quantifier-congruence derivation, not by treating double negations as identical numerical codes.

Applying this lemma to $\Gamma^-(B)$ gives

$$\mathcal{C}_- \vdash B \to U(N(B), B). \tag{1}$$

The binder in (1) is exactly $z = \mathrm{code}(N(B))$, not $z = \mathrm{code}(B)$. No relabeling or alpha-conversion of that binder has been asserted. Since $B < N(B)$, the chosen binder is still larger than every variable label appearing in B.

8.2 The two places in the inherited PA calibration

The supplied diagonal calibration replaces primitive equality vacuity in the original PA196 refutation compiler through two interfaces.

Reverse membership rewriting. In that argument the required formula is $C = E(a,b) \to (\chi(b) \to \chi(a))$, and the separator property is $\varphi = N(C)$ with separator label $c = \mathrm{code}(\varphi)$. The needed diagonal clause is exactly $\varphi \to U(c,\varphi) = \Gamma^-(C)$, which is an axiom of $\mathcal{C}_-$. Hence the negative-property specialization, pointed separation, label-collision cases, and two recovered reverse rewrites are unchanged. Membership vacuity is retained with its original guard; it has not been restricted to numerically large binders.

Neighborhood equality. For $B = E(a,b)$ choose

$$z_0 = \mathrm{code}(N(E(a,b))).$$

The positive-from-negative lemma gives $E(a,b) \to U(z_0, E(a,b))$. Lifting the two-way membership rewrites at z_0 then yields

$$E(a,b) \to R(a,b,z_0),$$

where

$$R(a,b,z) = U\big(z, M(a,z) \leftrightarrow M(b,z)\big) \wedge U\big(z, M(z,a) \leftrightarrow M(z,b)\big).$$

The freshness conditions hold because $z_0 > E(a,b) > a,b$. For any requested $z \neq a,b$, the existing membership-only binder-transport proof sends $R(a,b,z_0)$ to $R(a,b,z)$; if $z = z_0$ use the identity proof. That transport uses membership vacuity, the witness head and the already recovered rewrites, and needs no primitive equality vacuity. The subsequent neighborhood-witness interface is obtained by the same existential-monotonicity argument as before, using this updated neighborhood implication.

These are explicit replacements of the two diagonal-dependent calibration templates. They suffice to reuse the supplied PA196 refutation compiler at its stated rewriting/neighborhood/witness interface. Conversely every $\mathcal{C}_-$ axiom belongs to the original sound separate-head calculus. Thus, at the same inherited calibration boundary as the main report, there are primitive-recursive refutation translations with PA-provable correctness, and

$$\mathrm{PA} \vdash \left(\exists d\ \mathrm{Prf}_{\mathcal{C}_-}(d,1) \leftrightarrow \neg\,\mathrm{Con}(\mathrm{PA})\right).$$

The constructions use fixed finite proof templates, numerical constructor operations, a decidable binder-equality split and recursion on the input proof. No uniform arithmetic truth predicate, reflection principle, or unbounded search for fresh names is required. This does not claim that $\mathcal{C}_-$ derives every original $\Gamma(B)$ with its original binder B; the established result is the displayed refutation calibration.

8.3 Physical cost and the separate positive-offset experiment

The direct emitter for $\Gamma(N(D))$ uses three copies of the four-token expression `copyD, push, incT, cons`, followed by `cons, incT, incT, cons`. It therefore has sixteen tokens in place of the seven-token ordinary diagonal head. Nine direct insertions in the fixed A, D and Q grammar layouts, using each of A, D and H as the source parameter, had optimistic state counts at least 200. These profiles use the cached-MP, unsymmetric cache-test controller underlying those layouts, not the accepted DHS controller. These are compilation-cost screens, not accepted machines.

A separate simple schema emits $\Gamma(D+1)$. It covers all positive codes, while the omitted $\Gamma(0)$ follows from the truth theorem, generalization at zero, and propositional weakening. Consequently this positive-offset variant has a direct proof translation for every original diagonal instance. The recorded 72-case screen concerned $\Gamma(R+k)$, with $R \in \{A, D, H\}$, $1 \leq k \leq 8$, in the same three fixed grammar layouts; its minimum optimistic count was 198. Only $k = 1$ has been justified here by the preceding full-schema argument; other offsets in that screen were exploratory compilation cases.

Both screens subtract a maximum matching of eligible contextual pairs from the raw control count; such a matching may still fail simultaneous compatibility. They do not certify all source outcomes, blank startup, or any global minimum over machines. The data are retained in `supplement/negative_only_screen.json` and `supplement/positive_offset_screen.json`. The logical simplification is proved, but no accepted machine below 194 states follows from it.

The bundled standard-library replay script `supplement/semantic_screens.py` recomputes these 81 profiles and the 105 earlier controller/radix crosses from three frozen grammar inputs, using the pinned construction compiler. All 186 recorded profile counts were reproduced; the controller/radix cross minimum was 194.

9 Reproduction, artifact map and attribution

The package uses Python's standard library. Run without `-O` or `PYTHONOPTIMIZE`: assertions form part of the certificate checks. From the package root,

```
python3 -B verify_all.py
```

regenerates the construction, fresh zero relation, complete literal-tape contracts, exact startup, free-term command checks, novelty comparison and all-start obstruction. Stored receipts report the completed runs; the replay does not rerun heuristic discovery searches. A checkout of the pinned repository is needed only to regenerate the full corpus census, not to rebuild or verify the delivered candidate.

The optional semantic cost screens use the same standard-library runtime:

```
python3 -B supplement/semantic_screens.py
```

They reproduce all 186 stored profile counts. The independent local and grammar-search bundle also requires a C++17 compiler; see `search/local_and_grammar/README.md`. Its timed searches record proposal slots rather than distinct machines, and processor speed affects the number of proposals reached within each fixed budget.

To rebuild this report, run

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

Location	Contents
`report/`	Main TeX, compiled PDF, proof and generated table fragments.

Location	Contents
`machine/`	Definitive dense/native tables, raw parent, labels, source sites, ordinary/specialized instructions, fold list.
`construction/`, `inputs/`	Pinned compiler dependencies and parent material required for an offline rebuild.
`proof/`	Free-term emitter and all-variable command checks.
`verification/`	Independent literal-tape/source checker and credited symbolic-engine dependencies.
`evidence/`	Fresh build, command, source-invariant, full physical and start-up receipts.
`comparison/`	51-delivery ledger, full fresh census, 34 frontier representatives, exact novelty checker and independent review.
`search/`	Bounded search sources/receipts and the exact append-zero all-start obstruction.
`supplement/`	Restricted diagonal-schema proof, frozen grammar inputs, optimistic compilation-cost data and portable replay.
`references/`	Inherited ordinary PA calibration texts, original attributions and provenance.
`PROVENANCE.json`	Pinned repository, imported-file identities and artifact roles.
`SHA256SUMS`	Checksums of delivered files other than the checksum file itself.

The construction depends substantially on Vladimir Reshetnikov's ProveIt research packets, James Chang's PA196 development and the earlier register-engine work credited there. The runtime and logical machinery are inherited where identified. License notices and original dependency names are preserved. The new result is the DHS combination, its exact proof interface and fresh certificates, the current comparison and the all-start obstruction, and the restricted-schema refutation argument; it is not claimed as a new foundational proof calculus. The user's repository has not been modified.

References

[1] V. Reshetnikov, *ProveIt*, pinned commit `3cc7fbf75ba111a0da696ef2f6001dd528891bd5`, https://github.com/VladimirReshetnikov/ProveIt/tree/3cc7fbf75ba111a0da696ef2f6001dd528891bd5.

[2] *A 194-state Turing machine calibrated to the consistency of PA*, packet `pa194_conpa_2026-11-13`, in the pinned PA research documents.

[3] *PA194J and PA194D*, packet `pa194j_pa194d_conpa_2026-11-13`; especially `report/supplemental_gammad.tex` and `logical_emitter/rebuild.py`.

[4] *PA194HS*, incoming packet `pa194hs_conpa_2026-11-13.zip`; especially `report/proof_hybrid.tex`.

[5] *PA194NS*, incoming packet `pa194ns_conpa_2026-11-13.zip`; a duplicate dense HS table with its own report.

[6] *PA195*, packet `pa195_conpa_2026-11-13`; ordinary PA calibration, retained in `references/`.

[7] J. Chang, *PA196*, imported snapshot in `Computability/BusyBeaver/Research/PA/jamesyc-upstream/pa196`; `M6/Native.lean`, `M6/Runtime.lean`, `PA196/Program.lean`, and the source/calculus proofs.

[8] *PA consistency research*, packet `pa_consistency_research_2026-11-13_2`, `engine_search/README.md` and the rejected append-zero initializer trace.

[9] ProveIt, `Computability/BusyBeaver/Research/ZFC`, in particular the no-push and zero-spare transfer research and startup invariants.

[10] ProveIt, `Logic/PrimitiveRecursiveArithmetic/ConciseIPRA`, explicit compact proof systems and proof-translation methodology.

A Every supplied delivery

No.	Delivery	Reported contribution
1	`pa_unary_power_cover`	Unary power covering at PA strength (report of 27 October 2026). The U208 packets carry it as a baseline.
2	`U208_ConPA_Conditional_Context_2026-10-29`	U208 by a conditional tape-context reduction. States a 208-state candidate `machine/u208.tm`.
3	`U208_ConPA_Entry_Context_2026-10-29`	U208 by eliminating a state using entry context. States a 208-state machine `machine/u208.tm`.
4	`U208_ConPA_History_2026-10-29`	U208 by a finite-history decoder fold. States a 208-state candidate `machine/u208.tm`.
5	`U208_ConPA_2026-10-29`	U208 by a three-cell fold. States a 208-state candidate `machine/u208.tm`.
6	`u208_research_2026-10-29`	U208 research release. States two 208-state machines, `machine/u208.tm` and `machine/variant_plain/u208.tm`.
7	`Below_U208_Research_2026-10-29`	Research toward fewer than 208 states. States that no machine with 207 states or fewer was obtained.
8	`u208_comparison_research_2026-10-29`	Comparison of the 208-state proposals and a search below 208. States that the target was not achieved.
9	`ConPA_below208_research_2026-10-29`	Comparative audit and reduction search below 208. States that it does not establish a 207-state machine.
10	`pa_consistency_research_2026-10-29`	Audit and new source reductions. States that no validated machine below 208 states was obtained.
11	`pa208_research_2026-10-29`	Comparison and research below 208 states. States that it does not establish a valid machine below 208.
12	`pa_tm_research_2026-11-13`	Comparison and reduction research. States a new 196-state variant with shorter source paths and nothing below 196.
13	`pa_consistency_research_2026-11-13`	Comparison and diagonal-vacuity research. States a shorter axiom program with a 199-state implementation and nothing below 196.
14	`conpa_tm_comparison_and_reduction_search`	Comparison and reduction search. States two experimental 196-state command variants and nothing below 196.
15	`pa196_research_2026-11-13`	PA196 comparison, source factorization and proof obligations. States a source tree of 157 internal nodes and nothing below 196.
16	`pa_consistency_research`	Research and verification packet. States that no machine below 196 states has been validated.
17	`pa_tm_comparison_and_reductions`	Comparison and new rewrites, a research snapshot.

No.	Delivery	Reported contribution
18	`pa_tm_extended_research_2026-11-13`	Comparison and refined reduction search. States that the smallest accepted count is 196.
19	`conpa_tm_comparison_and_search`	Comparison and a search below 196. States that none was obtained; files named `195` are rejected or diagnostic candidates.
20	`pa_consistency_reduction`	Comparison and reduction experiments. States two new 196-state variants and a shorter source program with a 203-state implementation.
21	`pa_consistency_comparison_2026_11_13`	Comparison and new reduction experiments. States that nothing below 196 was obtained.
22	`PA_consistency_TM_research_2026-11-13`	Comparison and new source reductions. States that no machine below 196 is claimed.
23	`pa_consistency_research_2026-11-13_2`	Comparison and search below 196. States two new layouts with 196 and 198 states and a command-level reduction.
24	`conpa_tm_study_2026-11-13`	Comparison and reduction study. States that nothing below 196 was obtained.
25	`pa_below196_research`	Comparison and reduction experiments. States that nothing below 196 was obtained.
26	`pa_consistency_comparison_and_new196_2026-11-13`	Comparison and reduction research. States that it contains no accepted machine below 196.
27	`pa_tm_research_2026-11-13_2`	Comparison and reduction investigation. States same-size work, failed candidates and nothing below 196.
28	`pa_consistency_cross_archive`	Cross-archive constructions and reduction search. States that it does not certify a 195-state machine.
29	`pa195_conpa_2026-11-13`	PA195. States a 195-state machine `pa195/pa195.tm` and an article with a PA proof of the equivalence.
30	`pa_consistency_reduction_audit_2026-11-14`	Comparison and reduction audit. States nothing below 196 and no new theorem.
31	`pa_consistency_research_2026-11-14`	Comparison and NEG18. States an 18-instruction encoded negation with a compiled machine of 197 states.
32	`PA195_ConPA_research_2026-11-14`	A 195-state machine. States the primary `machine_195S.tm` and a second table `machine_195C.tm`.
33	`pa195_followup_research`	Comparison and searches below 195. States a graph-distinct 195-state machine M195S8 and nothing below 195.
34	`pa194_conpa_2026-11-13`	A 194-state machine. States the table `construction/tables/pa194.tm` and a proof of the equivalence in PA.
35	`pa194_conpa_2026-11-13_2`	PA194. States a 194-state machine `pa194/pa194.tm` and an article with a PA proof of the equivalence.
36	`pa195_diagonal_combined_2026-11-13`	PA195DC. States a different 195-state machine.
37	`pa194C_conpa_2026-11-13_2`	PA194C. States a materially different 194-state machine, an article with a PA proof of the equivalence, and nothing below 194.
38	`pa194_new_designs_2026-11-13`	Two new 194-state machines, PA194C and PA194CS. States that nothing below 194 was obtained.
39	`pa194c_conpa`	PA194C, a cached-head 194-state machine. States that no machine below 194 was established.
40	`pa194c_conpa_2026-11-13_1`	PA194C. States a different 194-state machine and an article with a PA proof of the equivalence.
41	`pa194c_conpa_2026-11-13`	PA194C, a cached-head 194-state machine. States an article with a PA proof of the equivalence and nothing below 194.
42	`pa194_radix_family_2026-11-13`	New 194-state machines calibrated to Con(PA), a radix family.
43	`pa194j_pa194d_conpa_2026-11-13`	PA194J and PA194D. States two new 194-state machines.

No.	Delivery	Reported contribution
44	`pa194q_conpa_2026-11-13`	PA194Q and PA194CSZ. States two new 194-state machines and that no machine below 194 was established.
45	`PA194R_research_2026-11-14`	PA194R. States a different 194-state machine with a new decoder layout and that no machine below 194 was verified.
46	`pa194N_conpa_2026-11-13`	PA194N. States a new 194-state machine, an article with a PA proof of the equivalence, and nothing below 194.
47	`pa194zr_conpa_2026-11-13`	PA194ZR. States a new 194-state machine `machine/pa194zr.tm`, a PA proof of the equivalence, and nothing below 194.
48	`pa194hs_conpa_2026-11-13`	**Current incoming ZIP.** PA194H and PA194HS combine no-push MP and cached-head Shift; HS uses symmetric comparison. HS is byte-identical to the separately named PA194NS.
49	`pa194ns_conpa_2026-11-13`	**Current incoming ZIP.** PA194NS hybrid no-push and symmetric cached-head Shift; byte-identical to PA194HS. New report and command proof, not an additional graph.
50	`pa194n5_n6_conpa_2026-11-13`	**Current incoming ZIP.** PA194N5 and PA194N6 combine radix-5/6 constructors and no-push MP. Supplies JavaScript universal checker; report PDF was typeset directly, not compiled from TeX.
51	`pa194n6_research_2026-11-13`	**Current incoming ZIP.** PA194N6 with two universal symbolic engines, PA command proof and calibration sources; exact table duplicates PA194N6 in the two-machine packet.

B Every supplied 194-state graph class

Name	Graph prefix	Category	Distinction / status
CS both-zero6	`76d35bfce3a6`	auxiliary	Known-zero X decrement at both source26 and final emitter exit; research composition.
CSZ-A3	`ea9f5da94795`	auxiliary	CSZ plus comparator-return register3 and final-emitter register3 decrements.
CSZ-A5	`bb730c3c718a`	auxiliary	CSZ plus comparator-return register3 and final-emitter register5 decrements.
CSZ-A6	`3b8e1d7018aa`	auxiliary	CSZ plus comparator-return register3 and final-emitter register6 decrements.
Cached I-tree zero-jump seed	`6f9dff0d3416`	auxiliary	Alternate decoder plus cached controller and entry zero jump; search basis.
Gamma(D)-CSZA	`64c88ee1c122`	auxiliary	Supplementary Gamma(D) emitter plus known-zero changes.
MPH	`5e4f51fc54c4`	auxiliary	MP result cached in H while retaining original Shift; physical search input.
PA194CZ	`080a545a20ef`	auxiliary	Cached controller; entry known-zero decrement of register 6.
PA194altC	`cf97d43185a6`	auxiliary	Cached controller on alternate 68-reader decoder.
Rejected CS node1-X	`dc936f437ad4`	rejected	Native103/read1 becomes dec6; exact blank run halts at1361.
Rejected I-tree node1-X	`aef95573862d`	rejected	Known-zero source rewrite fails blank bootstrap; exact halt at1361.
Rejected ZR node1-X	`2b53937097b9`	rejected	Register6 comparator-return variant; exact blank run halts at1361.

Name	Graph prefix	Category	Distinction / status
Rejected seven-zero scanner	`d2f0248d6556`	rejected	Decoder local folds pass, but source-address scanner violates runtime contract.
M4R	`a1109e46b45a`	reported construction	Reassociated coefficient4 constructor decoder; original resolved source graph.
M5	`debfb8a49978`	reported construction	Formula constructor coefficient5; explicit variable and proof recoding.
M6	`41d603308985`	reported construction	Formula constructor coefficient6; explicit variable and proof recoding.
PA194	`47e4859f85d4`	reported construction	Separate diagonal head, 67-reader decoder with seven folds, original controller.
PA194-68/8	`fcde56a35bab`	reported construction	Same separate-head source, reassociated 68-reader decoder with eight folds.
PA194-AD	`387b30b4e321`	reported construction	PA194 with physical A/D register exchange.
PA194A	`37d169082cbe`	reported construction	CS entry jump at source26 becomes decrement of known-zero register A.
PA194C	`9d787d29cb1c`	reported construction	MP result remains cached in H; Shift tests H=1; original ten-entry logical interface recompiled.
PA194CS	`90bca0127515`	reported construction	Symmetric Shift guard; cached MP controller; contextual controller fold.
PA194CSZ = PA194X	`b600465cb6d9`	reported construction	CS entry jump at source26 becomes decrement X; two headline names denote identical dense bytes.
PA194D	`c4ec894bec54`	reported construction	Separate diagonal axiom uses Gamma(D), new proof selector, 69 readers and nine folds.
PA194F	`bcdc219de3c4`	reported construction	CS final emitter jump becomes decrement of known-zero register X.
PA194H	`bc29bfd31443`	reported construction	No-push successful MP retains its conclusion in the visible top; Shift tests only cached head, using unsymmetric comparison.
PA194HS = PA194NS	`de41942ff26f`	reported construction	No-push successful MP with cached-head-only symmetric Shift. Two current incoming packets have the same exact dense table.
PA194J	`aa6c94af87a9`	reported construction	Ten source jumps become known-zero decrements; three physical rows change.
PA194N	`91e2e377b55f`	reported construction	Successful MP drops dummy-zero push, retains original Shift; revised command proof compiler.
PA194N5	`be8cbd683d7c`	reported construction	Formula coefficient 5 plus no-push successful MP; original top-plus-cache Shift condition.
PA194N6	`dba449969fa4`	reported construction	Formula coefficient 6 plus no-push successful MP; exact dense table repeated in two incoming packets.
PA194Q	`ccf29d739b36`	reported construction	Quantifier-distribution curried premises exchanged; K/S proof translation.
PA194R	`aea47aaf7ffa`	reported construction	Reassociated decoder of PA194F; 1111 source paths changed, same resolved source graph.
PA194ZR	`18e996382b72`	reported construction	Eight comparator returns and final emitter return become known-zero register3 decrements.

Graph prefixes abbreviate the full class hashes in `comparison/frontier194_annotated.json`, which also records source locators and exact byte digests. “Reported construction” describes the supplied evidence, not fresh validation of every machine.

C Complete transition table of PA194DHS

Each row gives a working state and its two actions. An action wDq means write w, move $D \in \{L, R\}$, and enter q. Start is 0; halt is 194. The machine-readable five-column file is authoritative and is bound by its SHA-256 in Section 1.

State	Read 0	Read 1	State	Read 0	Read 1
0	1R1	1R2	97	0R112	1R89
1	1R3	1R1	98	0R112	1R104
2	0R0	1R2	99	0R113	0L14
3	0R4	0R1	100	0R114	1L66
4	1R5	1L6	101	0R115	1R116
5	1L7	1L8	102	0R117	1R102
6	0L7	1L6	103	0R14	1R103
7	0L9	1L6	104	0L109	1L33
8	0L10	0L11	105	0R67	1R118
9	1R12	0L9	106	0R119	1L120
10	0L9	0L10	107	0R121	1R122
11	1L8	1L11	108	0R46	0L14
12	0R12	0R13	109	0R85	1R94
13	0L14	0R13	110	0R109	1L89
14	1L15	0L15	111	0R123	1R123
15	0L15	1L16	112	0L60	0R82
16	0L17	1L16	113	0R62	0R117
17	0L18	1L16	114	0R55	0R117
18	0L19	1L16	115	0L80	1R124
19	0L20	1L16	116	0R125	1R126
20	0L21	1L16	117	0R127	1R117
21	0L22	1L16	118	0R47	1R84
22	0L23	1L16	119	0R55	0R127
23	0R23	1R24	120	0R35	1R33
24	0R25	1R26	121	0R93	1R108
25	0R27	1R28	122	0R74	1R73
26	0R29	1R30	123	0R46	1R46
27	0R31	1R194	124	0R128	1R129
28	0L32	1R33	125	0R32	1R64
29	0R34	1R35	126	0R31	1R92
30	0L33	1L36	127	0R63	1R127
31	0R37	1R38	128	0R130	1R131
32	0R39	1R40	129	0R132	1R133
33	0R41	1R42	130	0R134	1R135
34	0R43	1R44	131	0R136	1R137
35	0R45	1R36	132	0R46	1R138
36	0R46	1R47	133	0R139	1R140
37	0R48	1R41	134	0R91	1R120
38	0R49	1R50	135	0R141	1R123
39	0R51	1R52	136	0R142	1R143
40	0R53	1R54	137	0R144	1R145
41	0R55	0R56	138	0L137	1R146
42	0R57	0L14	139	0L147	1R40
43	0R58	1R59	140	0R148	1R149
44	0R60	1R49	141	0L148	1R40
45	0R55	0R61	142	0R28	1R150
46	0R62	0R63	143	0R151	1R152

State	Read 0	Read 1	State	Read 0	Read 1
47	0L64	1L65	144	0R153	1R154
48	0R45	1R50	145	0R155	1R156
49	0L66	0L15	146	0R135	1R130
50	0R67	0L10	147	0R157	1R158
51	0R68	1R69	148	0R159	1R160
52	0R70	1R71	149	0R161	1R119
53	0R72	1R73	150	0R162	1R163
54	0R74	1R75	151	0R155	1R164
55	0L76	0R1	152	0R138	1R165
56	0R61	1R56	153	0R134	1R166
57	0L68	1L35	154	0R167	1R168
58	0R77	1R78	155	0L169	1R139
59	0R51	1R79	156	0R170	1R171
60	0L80	1R81	157	0R40	1R158
61	0R82	1R61	158	0L172	1L156
62	0L76	0R2	159	0R71	1R173
63	0R83	1R63	160	0R158	1R100
64	0R30	1R84	161	0R174	1R175
65	0R85	1R86	162	0R134	1R176
66	0R35	1R87	163	0R71	1L177
67	0R41	0L13	164	0R172	1R177
68	0R88	1R65	165	0R166	1R157
69	0R47	1R89	166	0R168	1R123
70	0R41	1R90	167	0L129	1R171
71	0R91	1R92	168	0L151	1L135
72	0R93	1L94	169	0R178	1R179
73	0R47	1R72	170	0R180	1R181
74	0R87	1R95	171	0R182	1R183
75	0R47	1R96	172	0R139	1R130
76	1R76	1L15	173	0R184	1R185
77	0R88	1R41	174	0R31	1R49
78	0R97	1R98	175	0R186	1R174
79	0R41	1R99	176	0R141	1R32
80	0R100	1R84	177	0R179	1R155
81	0R101	0L67	178	0R182	1R141
82	0R102	1R82	179	0R181	1R135
83	0R103	1R83	180	0R154	1R179
84	0R104	0L14	181	0R183	1R40
85	0R62	0R102	182	0R91	1R187
86	0R105	0L84	183	0L130	1R46
87	0R30	1R57	184	0R163	1R188
88	0L60	0R83	185	0R189	1R190
89	0R88	1L106	186	0R118	1R106
90	0R95	0L14	187	0R100	1R79
91	0R64	1R107	188	0R147	1R165
92	0R106	1R70	189	0R191	1R167
93	0L60	0R103	190	0R169	1R192
94	0L59	1R108	191	0R182	1R171
95	0L109	1R110	192	0R193	1R166
96	0R111	0L14	193	0R191	1R141